Modern Algebra 1

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Problem 1

Solution. Let G be a group with subgroup $H \leq G$ such that $[G : H] = 2$. We thus know H has two distinct left cosets, and equivalently has two distinct right cosets. Since H is a group and $e \in G$, we know one coset of H is $eH = H$. Let $a \in G$ such that $aH \neq H$. Thus, the two distinct cosets of H in G are H and aH. Since $eH = He$, it suffices to show $aH = Ha$. Recall from homework 5 that the set of distinct left cosets partition G, and thus $H \cup aH = G$. Conversely, $aH = G \setminus H$. Additionally, the set of right cosets partition G. Let $b \in G$ such that $Hb \neq He$. Thus, the set of distinct right cosets is $\{H, Hb\}$. Since the set of distinct right cosets partitions G, we know $H \cup Hb = G$. Conversely, $G \setminus H = Hb$. Since $G \setminus H = G \setminus H$, we have $aH = Hb$. It remains to be shown $Hb = Ha$.

Suppose for purpose of contradiction that $Hb \neq Ha$. Since there are only two cosets, it follows that $Ha = He$, and thus $a \in He$. Since $eH = He$, it follows that $a \in eH$, and thus $aH = eH$ by properties of cosets, which contradicts our definition. Therefore, $Hb = Ha$. Since we had $Hb = aH$, we have $aH = Ha$. Therefore, $H \triangleleft G$. ■

Problem 2

Solution. Let G be a nontrivial finite abelian group with order p^n for some prime p. By Lagrange's theorem, the order of every $g \in G$ must divide $|G|$, and thus $|g| \,|\, p^n$. Since the only divisors of a prime power p^n are prime powers p^m with $0 \leq m \leq n$, we know g must have $|g| = p^m$ for some m. In the case of $m = 0$, we have $|g| = 1$, and thus $g = e$.

Now we prove the converse. By the fundamental theorem of finite abelian groups,

$$G \cong \prod_{i=1}^{k} \mathbb{Z}_{p_i^{e_i}}$$

By properties of isomorphisms, if g corresponds to some element $(z_1, \ldots, z_n)$ in the direct product, then $|g| = |(z_1, \ldots, z_n)| = \operatorname{lcm}(|z_1|, \ldots, |z_n|)$. Since every element $g \in G$ has order p^a for some $a \in \mathbb{N}$, it follows that $\operatorname{lcm}(|z_1|, \ldots, |z_n|) = p^a$. It follows that for all i, $|z_i|$ divides p^a, and since the only divisors of p^a are p^b for $0 \leq b \leq a$, it follows that $|z_o|$ must be a power of p. Thus, all elements of $\mathbb{Z}_{p_i^{e_i}}$ are powers of p for all i. Notice that since each group $\mathbb{Z}_{p_i^{e_i}}$ is cyclic, it must have some element $z^* \in \mathbb{Z}_{p_i^{e_i}}$ such that $|z^*| = \left|\mathbb{Z}_{p_i^{e_i}}\right| = p_i^{e_i}$. Since we know the order of all elements of each cyclic group is p^b for some b, it follows that $p^b = p_i^{e_i}$. Therefore, $p_i = p$ for all i, and all cyclic groups have order p^{e_i}. It follows that

$$|G| = \left|\prod_{i=1}^{k} \mathbb{Z}_{p^{e_i}}\right| = \prod_{i=1}^{k} p^{e_i} = p^{\sum_{i=1}^{k} e_i}.$$

Therefore, all elements of G are of order p^n if and only if G is of order p^m. ■

Problem 3

Solution. Let G be an abelian group with even order, and let $H \leq G$ be a subgroup that contains all elements $g \in G$ with even order. By the fundamental theorem of finite abelian groups, we know

$$G \cong \prod_{i=1}^{n} \mathbb{Z}_{p_i^{e_i}}$$

for primes p_i. By this fact and our knowledge of direct products, it follows that

$$|G| \cong \prod_{i=1}^{n} p_i^{e_i}.$$

Since $|G|$ is even, we know 2 divides $|G|$. Since for any prime power $p_i^{e_i}$, its only divisors are prime powers p_i^m for $0 \leq m \leq e_i$, and since 2 is the only even prime, it follows that there must exist some i such that $p_i = 2$. Without loss of generality, let this i be 1.[1] It follows that

$$G \cong \mathbb{Z}_{2^{e_1}} \times \prod_{i=2}^{n} \mathbb{Z}_{p_i^{e_i}}.$$

Let ϕ be the isomorphism described by the fundamental theorem of finite abelian groups. Then for any $g \in G$, we can define

$$\phi(g) = (z_1, \ldots, z_n)\,,$$

where $z_1 \in \mathbb{Z}_{2^{e_1}}$, $z_2 \in \mathbb{Z}_{p_2^{e_e}}$, and so on. Note that $|g| = |\phi(g)|$ since ϕ is an isomorphism. We then have the following two cases.

Case 1: Suppose $z_1 \neq 0$. Since $z_1 \in \mathbb{Z}_{2^{e_1}} \setminus \{0\}$, we know $|z_1| > 1$, and thus $|z_1| \geq 2$. Since $|z_1|$ divides 2^{e_1}, there exists some $q \in \mathbb{Z}$ such that $q\,|z_1| = 2^{e_1}$. Since $e_1 \geq 1$, it follows that $2^{e_1 - 1} \in \mathbb{Z}$, and thus $2^{e_1-1} = \frac{q|z_1|}{2} \in \mathbb{Z}$, and thus 2 divides $|z_1|$.

Now consider the order of g, which is equivalent to $|\phi(g)| = |(z_1, \ldots, z_n)| = \operatorname{lcm}(|z_1|, \ldots, |z_n|)$. Note that the least common multiple is divided by all of its arguments, as it is multiples of all of them, and thus $|z_1|$ divides $|g|$. Since 2 divides $|z_1|$ and $|z_1|$ divides $|g|$, we know 2 divides $|g|$, and thus g has an even order. Therefore, $g \in H$.

Case 2: Suppose $z_1 = 0$. We then have

$$\phi(g) = (0, z_2, \ldots, z_n)\,.$$

From our results in case 1, we know if $\phi(g) = (z_1, \ldots, z_n)$ with $z_1 \neq 0$, then $g \in H$. It follows that $\phi(g) \in \phi(H)$ by definition of $\phi(H)$, and thus $(z_1, \ldots, z_n) \in \phi(H)$ for all $z_1 \neq 0$. It follows that

$$(1, \ldots, 1), (-1, z_2 - 1, \ldots, z_n - 1) \in \phi(H),$$

where $-1 \equiv 2^{e_1} - 1 \pmod{2^{e_1}}$. Since $\phi(H)$ is a group by our properties of isomorphisms, and thus closed, we have

$$(1, \ldots, 1) + (-1, z_2 - 1, \ldots, z_n - 1) = (0, z_2, \ldots, z_n) = \phi(g) \in \phi(H).$$

Therefore, $\phi(g) \in \phi(H)$. By definition of $\phi(H)$, this implies $g \in H$.

Therefore, $g \in H$ for all $g \in G$, and thus $G \subseteq H$. Since $H \leq G$, we have $H \subseteq G$, and thus $H = G$. ■

[1] Ordering the groups differently in the direct product is equivalent up to isomorphism, so we can take $i = 1$ without loss of generality here.

Problem 4

Solution. (a) Let $(a_1, a_2) \in G_1 \times G_2$ and let $(h_1, h_2) \in H_1 \times H_2$. It follows that $(a_1, a_2)(h_1, h_2) \in (a_1, a_2)(H_1 \times H_2)$. We have

$$(a_1, a_2)(h_1, h_2) = (a_1 h_1, a_2 h_2).$$

Since $a_1 \in G_1$, $h_1 \in H_1$, and $H_1 \triangleleft G_1$, we know there exists some $h_1' \in H_1$ such that $a_1 h_1 = h_1' a_1$. By the same logic, there exists $h_2' \in H_2$ such that $a_2 h_2 = h_2' a_2$. It follows that $(h_1', h_2') \in H_1 \times H_2$, and thus

$$(a_1 h_1, a_2 h_2) = \left(h_1' a_1, h_2' a_2\right) = \left(h_1', h_2'\right)(a_1, a_2) \in \left(H_1 \times H_2\right)(a_1, a_2).$$

Therefore, $(a_1, a_2)(H_1 \times H_2) \subseteq (H_1 \times H_2)(a_1, a_2)$. Applying the same logic in reverse yields $(a_1, a_2)(H_1 \times H_2) = (H_1 \times H_2)(a_1, a_2)$, and thus $H_1 \times H_2 \triangleleft G_1 \times G_2$.

(b) Let $A_1 = a_1 H_1$, and let $A_2 = a_2 H_2$ for $a_1 \in G_1$ and $a_2 \in G_2$. Note that $(a_1, a_2) \in G_1 \times G_2$. It follows that

$$A_1 \times A_2 = \left\{(a_1 h_1, a_2 h_2) \mid h_1 \in H_1, h_2 \in H_2\right\}.$$

Let $(a_1 h_1, a_2 h_2) \in A_1 \times A_2$. Notice that

$$(a_1 h_1, a_2 h_2) = (a_1, a_2)(h_1, h_2) \in (a_1, a_2)(H_1 \times H_2).$$

Therefore, $A_1 \times A_2 \subseteq (a_1, a_2)(H_1 \times H_2)$. Now let $(a_1, a_2)(h_1, h_2) \in (a_1, a_2)(H_1 \times H_2)$. It follows that

$$(a_1, a_2)(h_1, h_2) = (a_1 h_1, a_2 h_2) \in A_1 \times A_2.$$

Therefore, $A_1 \times A_2 = (a_1, a_2)(H_1 \times H_2)$. Note that since $(a_1, a_2) \in G$, then $(a_1, a_2)(H_1 \times H_2)$ is a coset of $H_1 \times H_2$ with coset representative (a_1, a_2).

(c) Let $(a_1, a_2) \in G_1 \times G_2$. Consider the coset $(a_1, a_2)(H_1 \times H_2)$. Let $(a_1, a_2)(h_1, h_2) \in (a_1, a_2)(H_1 \times H_2)$. It follows that

$$(a_1, a_2)(h_1, h_2) = (a_1 h_1, a_2 h_2).$$

Notice that $a_1 h_1 \in a_1 H_1$ and $a_2 h_2 \in a_2 H_2$. It follows that $(a_1, a_2)(H_1 \times H_2) \subseteq a_1 H_1 \times a_2 H_2$. Now we prove the converse. Let $(a_1 h_1, a_2 h_2) \in a_1 H_1 \times a_2 H_2$. It follows that

$$(a_1 h_1, a_2 h_2) = (a_1, a_2)(h_1, h_2) \in (a_1, a_2)(H_1 \times H_2).$$

Therefore, for any coset $(a_1, a_2)(H_1 \times H_2)$, we have

$$(a_1, a_2)(H_1 \times H_2) = a_1 H_1 \times a_2 H_2.$$

(d) Let $\phi : (G_1/H_1) \times (G_2/H_2) \to (G_1 \times G_2) / (H_1 \times H_2)$ be the map

$$(a_1 H_1, a_2 H_2) \mapsto (a_1, a_2)(H_1 \times H_2).$$

We must show ϕ is an isomorphism. Before we do that, we will list a few facts that are important to know.

- Since equality for ordered pairs is defined componentwise, $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.
- By part (c), $(a_1, a_2)(H_1 \times H_2) = a_1 H_1 \times a_2 H_2$ for all $(a_1, a_2) \in G_1 \times G_2$.

First, we show ϕ is well defined. Let $(a_1H_1, a_2H_2), (b_1H_1, b_2H_2) \in (G_1/H_1) \times (G_2/H_2)$ such that $(a_1H_1, a_2H_2) = (b_1H_1, b_2H_2)$. Notice that $a_1H_1 = b_1H_1$ and $a_2H_2 = b_2H_2$. It follows that

$$\begin{aligned}
\phi(a_1H_1, a_2H_2) &= (a_1, a_2)(H_1 \times H_2) \\
&= a_1H_1 \times a_2H_2 \\
\phi(b_1H_1, b_2H_2) &= (b_1, b_2)(H_1 \times H_2) \\
&= b_1H_1 \times b_2H_2
\end{aligned}$$

Since $a_1H_1 = b_1H_1$ and $a_2H_2 = b_2H_2$, we know

$$= a_1H_1 \times a_2H_2 = b_1H_1 \times b_2H_2.$$

Therefore,

$$\phi(a_1H_1, a_2H_2) = \phi(b_1H_1, b_2H_2).$$

Therefore, ϕ is well defined.

Now, we show ϕ is a homomorphism. Let definitions be as before. We have

$$\begin{aligned}
\phi\left(a_1H_1, a_2H_2\right)\phi\left(b_1H_1, b_2H_2\right) &= \left((a_1, a_2)(H_1 \times H_2)\right)\left((b_1, b_2)(H_1 \times H_2)\right) \\
&= (a_1, a_2)(b_1, b_2)(H_1 \times H_2) \\
&= (a_1b_1, a_2b_2)(H_1 \times H_2) \\
&= \phi(a_1b_1H_1, a_2b_2H_2).
\end{aligned}$$

Therefore, ϕ is a homomorphism. Now, we will show ϕ is injective. Let definitions be as before, and suppose $\phi(a_1H_1, a_2H_2) = \phi(b_1H_1, b_2H_2)$. It follows that $(a_1, a_2)(H_1 \times H_2) = (b_1, b_2)(H_1 \times H_2)$, and thus, $a_1H_1 \times a_2H_2 = b_1H_1 \times b_2H_2$. We must show $a_1H_1 = b_1H_1$ and $a_2H_2 = b_2H_2$. Since $a_1H_1 \times a_2H_2 = b_1H_1 \times b_2H_2$, for all $h_1 \in H_1$ and $h_2 \in H_2$, there exists $h_1' \in H_1$ and $h_2' \in H_2$ such that

$$(a_1h_1, a_2h_2) = (b_1h_1', b_2h_2').$$

Therefore, $a_1H_1 \subseteq b_1H_1$, and $a_2H_2 \subseteq b_2H_2$. Applying the same logic in reverse will show $b_1H_1 \subseteq a_1H_1$ and $b_2H_2 \subseteq a_2H_2$, and thus $(a_1H_1, a_2H_2) = (b_1H_1, b_2H_2)$. Therefore, ϕ is injective.

Finally, we show ϕ is surjective. Let $(a_1, a_2)(H_1 \times H_2) \in (G_1 \times G_2)/(H_1 \times H_2)$. It follows that $a_1 \in G_1$ and $a_2 \in G_2$, and thus $a_1H_1 \in G_1/H_1$ and $a_2H_2 \in G_2/H_2$. Therefore,

$$\phi(a_1H_1, a_2H_2) = (a_1, a_2)(H_1 \times H_2).$$

Therefore, ϕ is surjective, and is thus an isomorphism. Therefore,

$$(G_1/H_1) \times (G_2/H_2) \cong (G_1 \times G_2)/(H_1 \times H_2).$$

$\blacksquare$

Problem 5

Solution. (a) Let G be a group. Notice that for all $g \in G$, we know by definition that for all $z \in Z(G)$, $zg = gz$. It follows that $gZ(G) = Z(G)g$ for all $g \in G$, and thus $Z(G) \triangleleft G$.

(b) Suppose that $G/Z(G)$ is cyclic. It follows that there exists some generator $xZ(G)$ such that $G/Z(G) = \langle xZ(G) \rangle$. It follows that for all $g \in G$, there exists some integer m such that $gZ(G) = (xZ(G))^m = x^mZ(G)$. It follows that for all $z_1 \in Z(G)$ there exists some $z_2 \in Z(G)$

such that $gz_1 = x^m z_2$. Thus, $g = z_1^{-1} x^m z_2$. By the same logic, for any $h \in G$, there exist z_3, z_4, and an integer n such that $h = z_3^{-1} x^n z_4$. Since z_1, z_2, z_3, z_4 commute with all elements of G and x commutes with itself, it follows that

$$\begin{aligned} gh &= z_1^{-1} x^m z_2 z_3^{-1} x^n z_4 \\ &= z_3^{-1} z_4 x^m x^n z_1^{-1} z_2 \\ &= z_3^{-1} z_4 x^n x^m z_1^{-1} z_2 \\ &= z_3^{-1} x^n z_4^{-1} z_1^{-1} x^m z_2 \\ &= hg, \end{aligned}$$

for any $g, h \in G$. Therefore, G is abelian.

(c) Since $[G : Z(G)] = 4$, there exist 4 distinct cosets of $Z(G)$. Since up to isomorphism, there are only two groups of order 4, namely $\mathbb{Z}_4$ and $\mathbb{Z}_2^2$ (proof in homework 5), it suffices to show $G/Z(G)$ is not cyclic. Suppose for purpose of contradiction that $G/Z(G)$ is cyclic. By part (b), G is abelian. It follows that by definition, $Z(G) = G$. Recall that $[G : Z(G)] = \frac{|G|}{|Z(G)|}$. Since $Z(G) = G$, we have $[G : Z(G)] = \frac{|G|}{|G|} = 1$, contradicting the fact that $[G : Z(G)] = 4$. Thus, $Z(G)$ is not cyclic, and therefore must be isomorphic to $\mathbb{Z}_2^2$.

After rereading my proof, I realized this fact that $[G : H] = \frac{|G|}{|H|}$ only holds for finite groups, so I have done a stronger proof. Suppose again for purpose of contradiction that $Z(G)$ is cyclic, and notice again that $Z(G) = G$ by part (b). In the proof of Cayley's theorem, we showed $T_a : g \mapsto ag$ is a permutation of G, and by definition of a permutation it follows that $T_a(G) = \{ag \mid g \in G\} = G$ for any fixed $a \in G$. Notice that

$$T_a(G) = \{ag \mid g \in G\} = \{ag \mid g \in Z(G)\} = aZ(G),$$

since $G = Z(G)$. Since $a \in G$ is arbitrary, this representation constitutes all cosets of $Z(G)$, and thus all cosets are the same by this fact. Therefore, there only exists one coset, giving us the same contradiction as before. I will spend the remainder of the paper proving $g \mapsto ag$ is a permutation of G, since we didn't complete this part of the proof in class so I wasn't sure if you needed it.

For all $a \in G$, define $T_a : G \to G$ to be the map $g \mapsto ag$. First we show T_a is surjective. Let $g \in G$. Since G is a group and thus closed, $a^{-1}g \in G$. Notice that $T_a(a^{-1}g) = aa^{-1}g = g$. Therefore, $T_a(g)$ is surjective. Next, we show T_a is injective. Let $T_a(g) = T_a(h)$ for some $g, h \in G$. It follows that $ag = ah$. Since G is a group and thus contains inverses, it follows that $a^{-1}ag = a^{-1}ah$, and thus $g = h$. Therefore, T_a is a bijection, and since it is a map $G \to G$, T_a is a permutation of G. ■